Kempower challenge: The Self-Operating Charging Network

LahtiES Energy Tech Hackathon · Thu 5.11. – Sat 7.11.2026 · LUT Lahti / CrazyTown

> **"If humans were no longer allowed to continuously operate and maintain a charging network, how would you redesign it?"**

About Kempower

Kempower designs and manufactures DC fast charging solutions for electric vehicles: cars, buses, trucks, machines and boats. The company was founded in Lahti in 2017, and its chargers are in use across Europe and North America. More at kempower.com.

Background

A charging network is an energy system. Every site has a grid connection with a power limit, electricity prices change every 15 minutes, buses and trucks must leave on time, drivers do not want to queue, and chargers fail. Today people balance all of this: they monitor, schedule, re-plan, plug and re-plug vehicles, reset chargers, send technicians, bring new sites online and make promises to customers.

Whose network is it?

Many actors share a charging network, and each sees the problem differently:

- the **charger manufacturer** that builds the equipment and often services it
- the **charge point operator** that runs public charging and sells it to drivers
- the **site owner**: a bus company, a logistics company, a shop, a port
- the **fleet operator** whose vehicles must be charged and on the road in time
- the **energy company and grid operator** that supply power and set the limits
- the **drivers**

Pick the perspective you design from, or combine several, and say which one you chose. The best answer may change who does what.

Your mission

Design the intelligence that runs a charging network without continuous human involvement, and show it working.

The challenge has three parts:

1. **The energy and control core.** How is power shared between chargers, sites and hours when grid limits, prices and vehicle schedules pull in different directions? How does the network notice a problem and recover without a person? Where does the intelligence live: in the charger, at the site, in the cloud or in an external system, and what happens when the connection is lost?

2. **Show it.** A prototype, simulation or model on the dataset we provide. Run the network through **Friday 13.11.2026** (the scenario day in the data) with no human intervention, including at least one failure and one electricity price event. The level is your choice, from a spreadsheet model to working code.
3. **The commercial side.** When no one operates the network, who is the customer and what do they buy? Propose a business model and a one-page service agreement: what is promised, what the system decides on its own, and who is accountable when it decides wrong.

The scope is open on purpose. People do many different jobs in a charging network today, and the data covers many of them. Choose where to take the human out first. Going deep on one area is better than touching everything lightly.

What you deliver on Saturday 7.11.

- **A 10-minute pitch** built around one customer story set in 2029.
- **The demonstration** (part 2).
- **The one-page service agreement** (part 3).

What we provide

- **A synthetic dataset** (`kempower-challenge-data/`, CSV, opens in Excel). A fictional network of 100 charging sites in Finland, Sweden, Norway, Denmark and Germany, with depots owned by fleet companies and public sites run by an operator, two weeks at 15-minute resolution, plus five close-up sites in a fictional Finnish town with every charger, vehicle, fault and service visit. Start with `README.md` in the data folder; every column is described in `DATA_DICTIONARY.md`.
- **Public sources** for real electricity, grid, traffic and product data (`PUBLIC_SOURCES.md`).
- **Mentors from Kempower:**
    - Thursday morning: kickoff and Q&A with Mats Mattsson and Panu Vartiainen
    - Thursday and Friday: Panu Vartiainen
    - Saturday: Mats Mattsson, presentations and judging

The data is synthetic: it describes no real company, customer or product. Mentors answer questions about the data and the charging industry, but they will not suggest solutions and cannot share internal company information. The ideas are yours.

How we evaluate

Criterion	Weight	What the judges look for
Degree of autonomy	25 %	How much of the network's normal operation runs without a person, and how convincingly you show it. Where humans still stay in the loop, and why.
Energy and grid impact	20 %	What your design does to energy cost, peak power, grid limits and the use of available energy, shown with numbers from the data.
Technical credibility	20 %	Does it work on the data, including the failure and the price event? Are the assumptions stated and sensible? What happens when something goes wrong?
Commercial viability	20 %	Would a customer pay for it? Is the service agreement clear about promises, decisions and accountability?

Criterion	Weight	What the judges look for
Clarity	15 %	Is the 2029 customer story easy to follow? Can a non-specialist understand what changed and why it matters?

The baseline numbers in `baseline_kpis.csv` and `network_kpis.csv` show how the network performs with people in the loop today. Use them as a reference point when you show your results.

Practical

- Any tools are fine: Excel, Python, simulation tools, slides, mock-ups.
- The event rules in the LahtiES info package apply, including the rules on intellectual property.
- Questions before the event: ask LahtiES, who will pass them on.

About this data

Everything in this dataset was generated by a computer program for this hackathon. The sites, locations, vehicles, customers, contracts, prices, costs, faults, firmware versions and events are invented. Nothing comes from Kempower's or anyone else's systems, and nothing describes Kempower's products, prices, customers, service terms or business. Electricity prices, weather and traffic are synthetic as well; they only resemble real patterns. Any resemblance to real sites, companies or incidents is coincidental. For real figures, use the public sources listed in `PUBLIC_SOURCES.md` in the data folder.

Good luck, and have fun!